

# Spring Data JPA with Spring Boot, Hibernate

## Spring Data JPA - Quick Example

```

:: Spring Boot :: (v4.1.0)

02-07-26 11:46:27.538 restartedMain INFO c.c.o.OrmLearnApplication logStarting 54 Starting OrmLearnApplication using Java 25.0.3 with PID 27748 (C:
02-07-26 11:46:27.540 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 55 Running with Spring Boot v4.1.0, Spring v7.0.8
02-07-26 11:46:27.541 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 652 No active profile set, falling back to 1 default profile: "default"
02-07-26 11:46:27.585 restartedMain INFO ertyDefaultsPostProcessor log 69 Devtools property defaults active! Set 'spring.devtools.add-props' to 'true' to enable
02-07-26 11:46:27.917 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 145 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
02-07-26 11:46:27.949 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 213 Finished Spring Data repository scanning in 26 ms. Found 1 JPA repository
02-07-26 11:46:28.198 restartedMain INFO org.hibernate.orm.jpa logPersistenceUnitInformation 102 HHH0008540: Processing PersistenceUnitInfo [name: default]
02-07-26 11:46:28.236 restartedMain INFO org.hibernate.orm.core logVersion 41 HHH000001: Hibernate ORM core version 7.4.1.Final
02-07-26 11:46:28.532 restartedMain INFO SpringPersistenceUnitInfo addTransformer 129 No LoadTimeWeaver setup: ignoring JPA class transformer
02-07-26 11:46:28.559 restartedMain INFO c.z.h.HikariDataSource getConnection 109 HikariPool-1 - Starting...
02-07-26 11:46:28.886 restartedMain INFO c.z.h.pool.HikariPool checkFailFast 580 HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@10000000
02-07-26 11:46:28.890 restartedMain INFO c.z.h.HikariDataSource getConnection 122 HikariPool-1 - Start completed.
02-07-26 11:46:29.029 restartedMain INFO o.h.o.connections.pooling logConnectionInfo 179 HHH10001005: Database info:
    Database JDBC URL [jdbc:mysql://localhost:3306/ormlearn]
    Database driver: MySQL Connector/J
    Database dialect: MySQLDialect
    Database version: 8.0.45
    Default catalog/schema: ormlearn/undefined
    Autocommit mode: undefined/unknown
    Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
    JDBC fetch size: none
    Pool: DataSourceConnectionProvider
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown

02-07-26 11:46:29.718 restartedMain INFO org.hibernate.orm.core initiateService 53 HHH000489: No JTA platform available (set 'hibernate.transaction
02-07-26 11:46:29.862 restartedMain INFO nEntityManagerFactoryBean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for persistence unit 'default'
02-07-26 11:46:29.998 restartedMain INFO .q.QueryEnhancerFactories <clinit> 48 Hibernate is in classpath; If applicable, HQL parser will be used
02-07-26 11:46:30.475 restartedMain INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearnApplication in 3.286 seconds (process running for 3.286 seconds)
02-07-26 11:46:30.483 restartedMain INFO c.c.o.OrmLearnApplication main 23 Inside main
02-07-26 11:46:30.484 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 31 Start
02-07-26 11:46:30.700 restartedMain DEBUG org.hibernate.SQL logStatement 126 select c1_0.co_code,c1_0.co_name from country c1_0
02-07-26 11:46:30.754 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 33 countries=[Country{code='IN', name='India'}, Country{code='US', name='United States'}]
02-07-26 11:46:30.754 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 34 End
02-07-26 11:46:30.758 locationShutdownHook INFO nEntityManagerFactoryBean destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
02-07-26 11:46:30.760 locationShutdownHook INFO c.z.h.HikariDataSource close 349 HikariPool-1 - Shutdown initiated...
02-07-26 11:46:30.765 locationShutdownHook INFO c.z.h.HikariDataSource close 351 HikariPool-1 - Shutdown completed.

Process finished with exit code 0
```

## Difference between JPA, Hibernate and Spring Data JPA

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Run EmployeeCrudApplication x

Console Beans Health Mappings Environment

2026-07-03T14:00:57.575+05:30 INFO 29412 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-07-03T14:00:57.954+05:30 INFO 29412 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@10000000
2026-07-03T14:00:57.958+05:30 INFO 29412 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-07-03T14:00:58.109+05:30 INFO 29412 --- [main] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
    Database JDBC URL [jdbc:mysql://localhost:3306/employeeeb]
    Database driver: MySQL Connector/J
    Database dialect: MySQLDialect
    Database version: 8.0.45
    Default catalog/schema: employeeeb/undefined
    Autocommit mode: undefined/unknown
    Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
    JDBC fetch size: none
    Pool: DataSourceConnectionProvider
    Minimum pool size: undefined/unknown
    Maximum pool size: undefined/unknown

2026-07-03T14:00:58.774+05:30 INFO 29412 --- [main] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.transaction
Hibernate: create table employee (id integer not null auto-increment, department varchar(255), name varchar(255), primary key (id)) engine=InnoDB
2026-07-03T14:00:58.946+05:30 INFO 29412 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-07-03T14:00:59.042+05:30 INFO 29412 --- [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will be used
2026-07-03T14:00:59.178+05:30 WARN 29412 --- [main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database access may cause requests to hang or timeouts (timeout due to socket options)
2026-07-03T14:00:59.506+05:30 INFO 29412 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-03T14:00:59.510+05:30 INFO 29412 --- [main] c.c.e.EmployeeCrudApplication : Started EmployeeCrudApplication in 3.617 seconds (process running for 3.617 seconds)
Hibernate: insert into employee (department,name) values (?,?)
Saved Employee ID: 1
```